

WASP-36b

R-0996

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_180126 · created 2026-07-30T00:38:12+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458144.93 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.048
Transit depth (Rp/Rs)^2	2.56 +/- 1.53 [%]
Orbital Inclination (inc)	81.42 +/- 2.89
Airmass coefficient 1 (a1)	0.54 +/- 0.36
Airmass coefficient 2 (a2)	0.12 +/- 0.11
Scatter in the residuals of the lightcurve fit is	59.62 %
Best Comparison Star	None
Optimal Aperture	1.93
Optimal Annulus	5.75
Transit Duration (day)	0.057 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.16 $\pm$ 0.048 vs 0.1368 (deviation 0.18 σ)
Duration fit vs published	0.057 d vs 0.0758 d (deviation 0.22 σ)
Mid-time O-C	6.8 min
Depth SNR	1.2
Residual scatter	59.62 %

Light curve

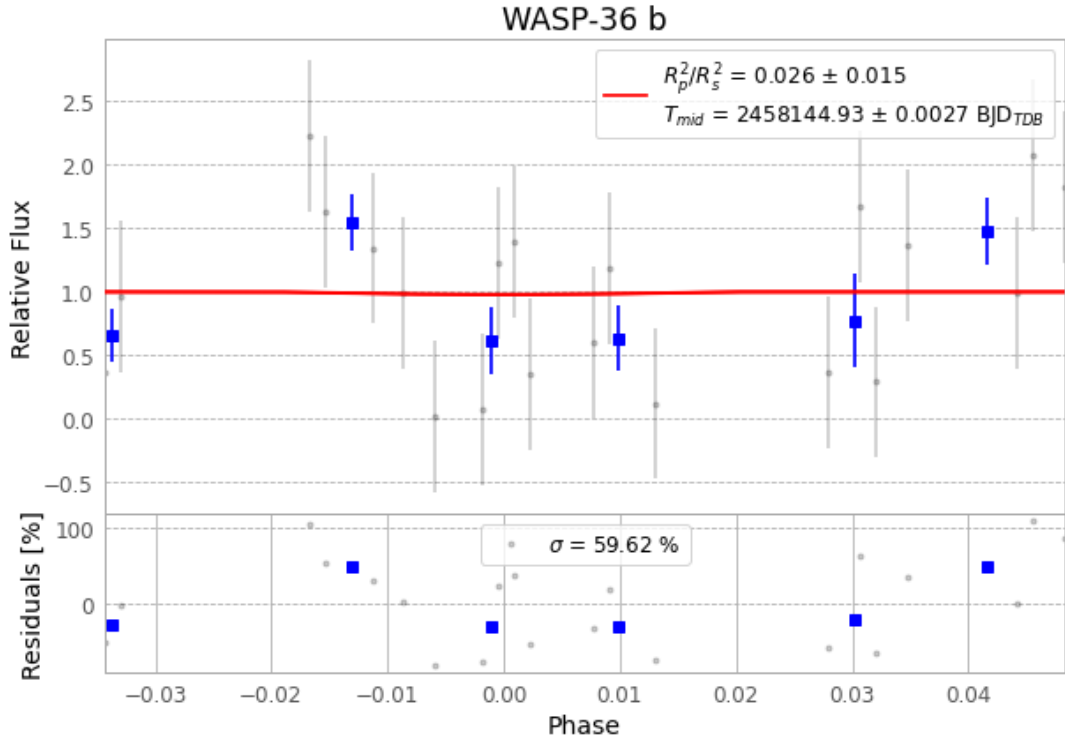


Figure 1 — FinalLightCurve_WASP-36 b_2018-01-26.png (SHA-256 60210e7b6b909cb0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [452, 463], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e5296704a97371bc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

62 FITS frames (41 MB), WASP-36180126085412.FITS → WASP-36180126115712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-180126.log (fda7d74cc79b...), FinalLightCurve_WASP-36 b_2018-01-26.png (60210e7b6b90...), FinalLightCurve_WASP-36 b_2018-01-26.csv (347e4fec64d...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 00:38Z.